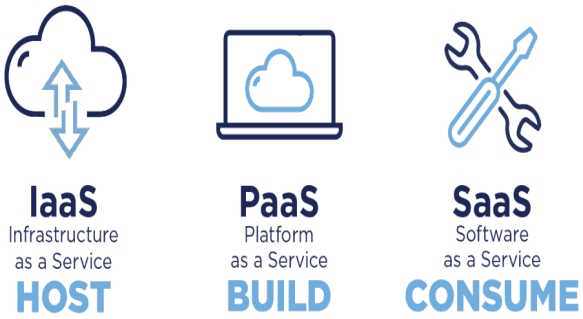
Chapter 3

Fundamental Concepts and Models



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Outlines

- Roles and Boundaries
- Cloud Characteristics
- Cloud Delivery Models
- Cloud Deployment Models

Roles and Boundaries

Roles

Roles

- Organizations and humans can assume different types of pre-defined roles depending on how they relate to and/or interact with a cloud and its hosted IT resources.
- Cloud Provider
- Cloud Consumer
- Cloud Service Owner
- Cloud Resource Administrator

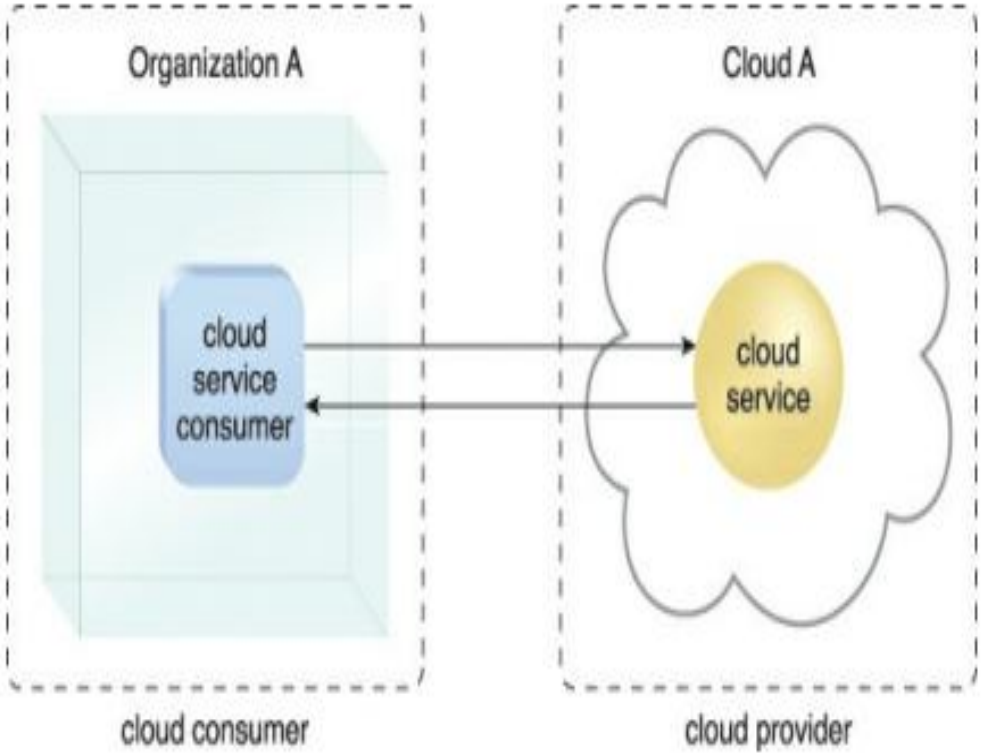
Are among the well classified roles

Cloud Provider

- The organization that provides cloud-based IT resources is the cloud provider.
- When assuming the role of cloud provider, an organization is responsible for making cloud services available to cloud consumers, as per agreed upon SLA guarantees.
- The cloud provider is further tasked with any required management and administrative duties to ensure the on-going operation of the overall cloud infrastructure.
- Cloud providers normally own the IT resources that are made available for lease by cloud consumers; however, some cloud providers also "resell" IT resources leased from other cloud providers

Cloud **Consumer**

- A cloud consumer is an organization (or a human) that has a formal contract or arrangement with a cloud provider to use IT resources made available by the cloud provider.
- Specifically, the cloud consumer uses a cloud service consumer to access a cloud service



Cloud Service Owner

- Person(organization) that legally owns a cloud service is called a cloud service owner.
- The cloud service owner can be the *cloud consumer*, or the *cloud provider* that owns the cloud within which the cloud service resides.
- *For example, either the cloud consumer of Cloud X or the cloud provider of Cloud X could own Cloud Service A*
- Note that a cloud consumer that owns a cloud service hosted by a third-party cloud does not necessarily need to be the user (or consumer) of the cloud service.
- Several cloud consumer organizations develop and deploy cloud services in clouds owned by other parties for the purpose of making the cloud services available to the general public.

Cloud Service Owner

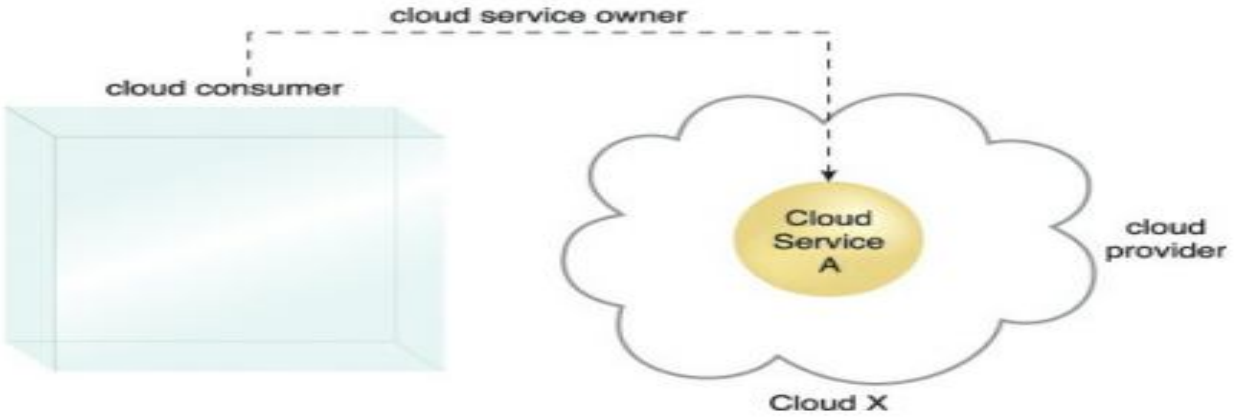


Figure A cloud consumer can be a cloud service owner when it deploys its own service in a cloud.



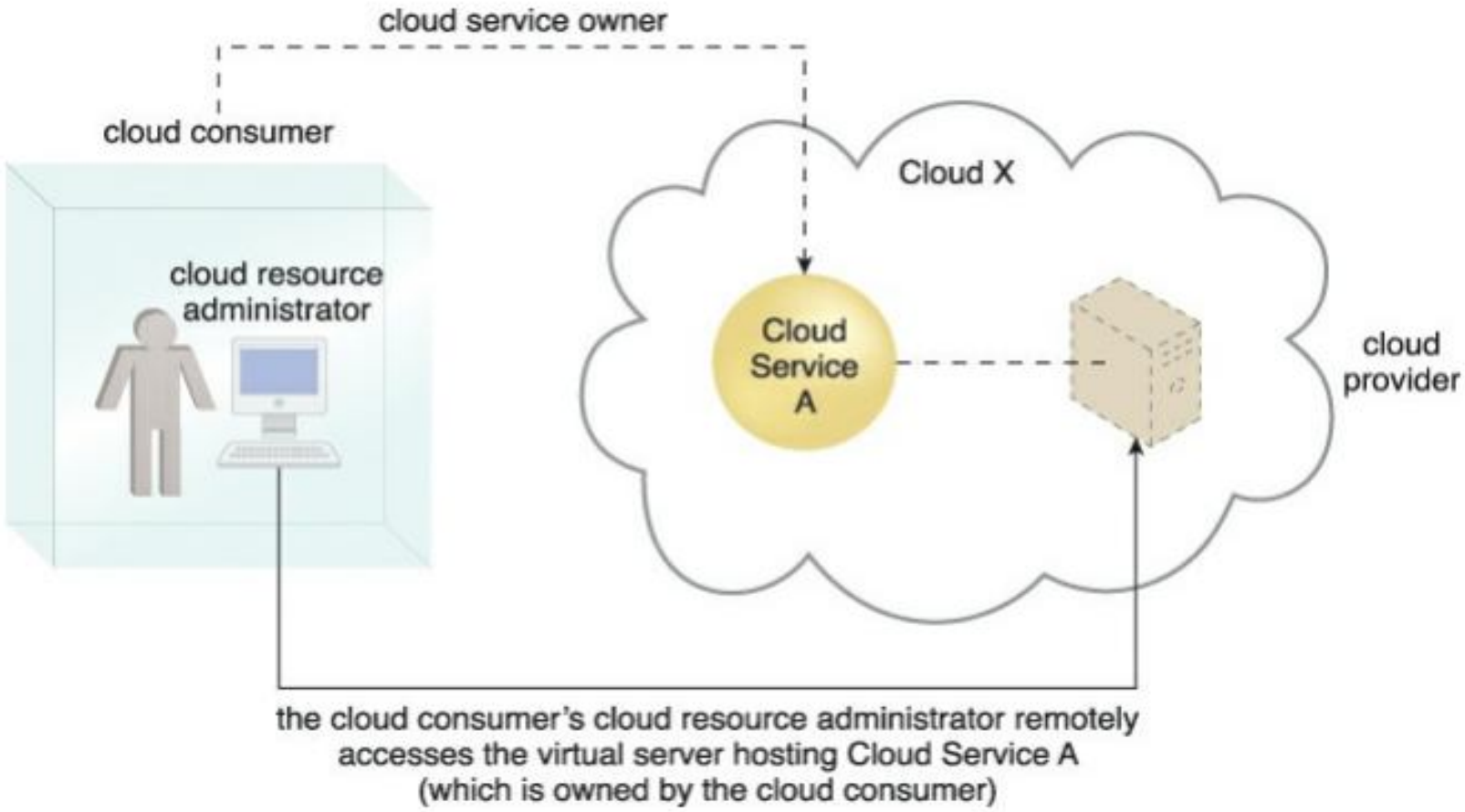
Figure A cloud provider becomes a cloud service owner if it deploys its own cloud service, typically for other cloud consumers to use.

Cloud Resource Administrator

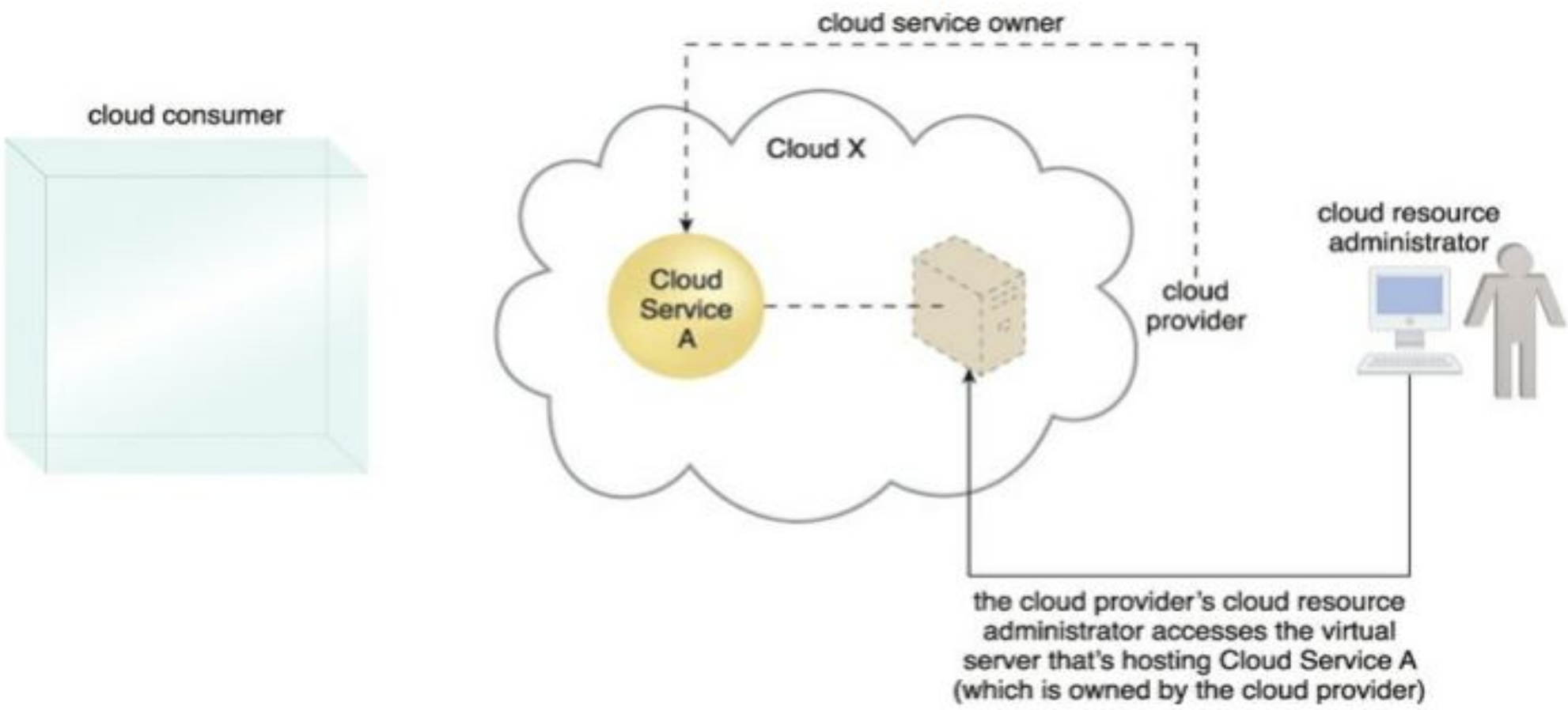
- A cloud resource administrator is the person or organization responsible for administering a cloud-based IT resource (including cloud services).
- Cloud resource administrator can be (or belong to) the cloud consumer or cloud provider of the cloud within which the cloud service resides.
- Alternatively, it can be (or belong to) a third-party organization contracted to administer the cloud-based IT resource.
- *For example, a cloud service owner can contract a cloud resource administrator to administer a cloud service*

Cloud Resource Administrator

- The reason a cloud resource administrator is not referred to as a "cloud service administrator" is because this role may be responsible for administering cloud-based IT resources that don't exist as cloud services.
- *For example, if the cloud resource administrator belongs to (or is contracted by) the cloud provider, IT resources not made remotely accessible may be administered by this role (and these types of IT resources are not classified as cloud services).*



"... A cloud resource administrator can be with a cloud consumer organization and administer remotely accessible IT resources that belong to the cloud consumer.



A cloud resource administrator can be with a cloud provider organization for which it can administer the cloud provider's internally and externally available IT resources.

Additional Roles

- The NIST Cloud Computing Reference Architecture defines the following supplementary roles:
  - ✓ Cloud Auditor
  - ✓Cloud Broker
  - ✓ Cloud Carrier

Cloud Auditor

- These are third-party (often accredited) organizations that conducts independent assessments of cloud environments.
- The typical responsibilities associated with this role include the evaluation of security controls, privacy impacts, and performance.
- The main purpose of the cloud auditor role is to provide an unbiased assessment (and possible endorsement) of a cloud environment to help strengthen the trust relationship between cloud consumers and cloud providers.

Cloud Broker

- This role is assumed by a party that assumes the responsibility of managing and negotiating the usage of cloud services between cloud consumers and cloud providers.
- Mediation services provided by cloud brokers include service intermediation, aggregation, and arbitrage.

Cloud Carrier

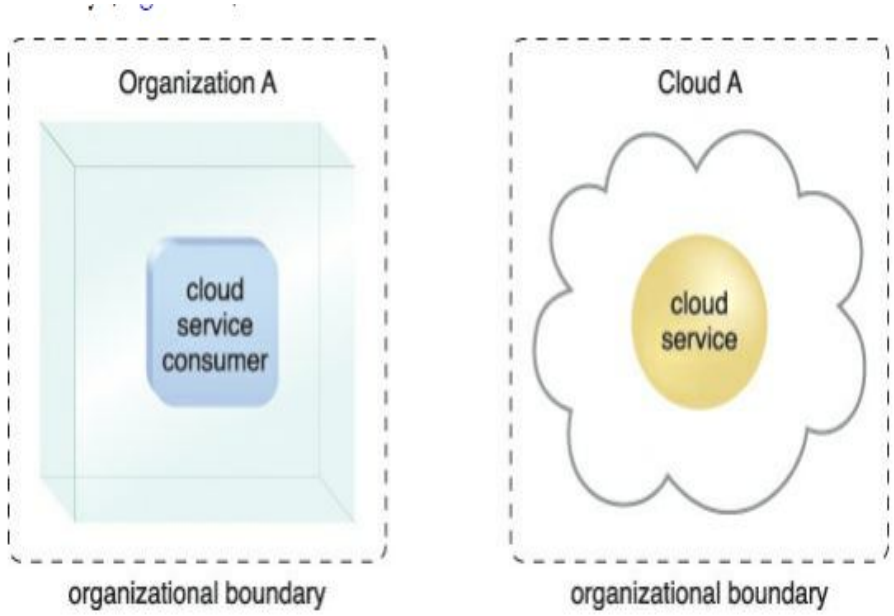
- The party responsible for providing the wire-level connectivity between cloud consumers and cloud providers assumes the role of the cloud carrier.
- This role is often assumed by network and telecommunication providers.

Roles and Boundaries



Organizational Boundary

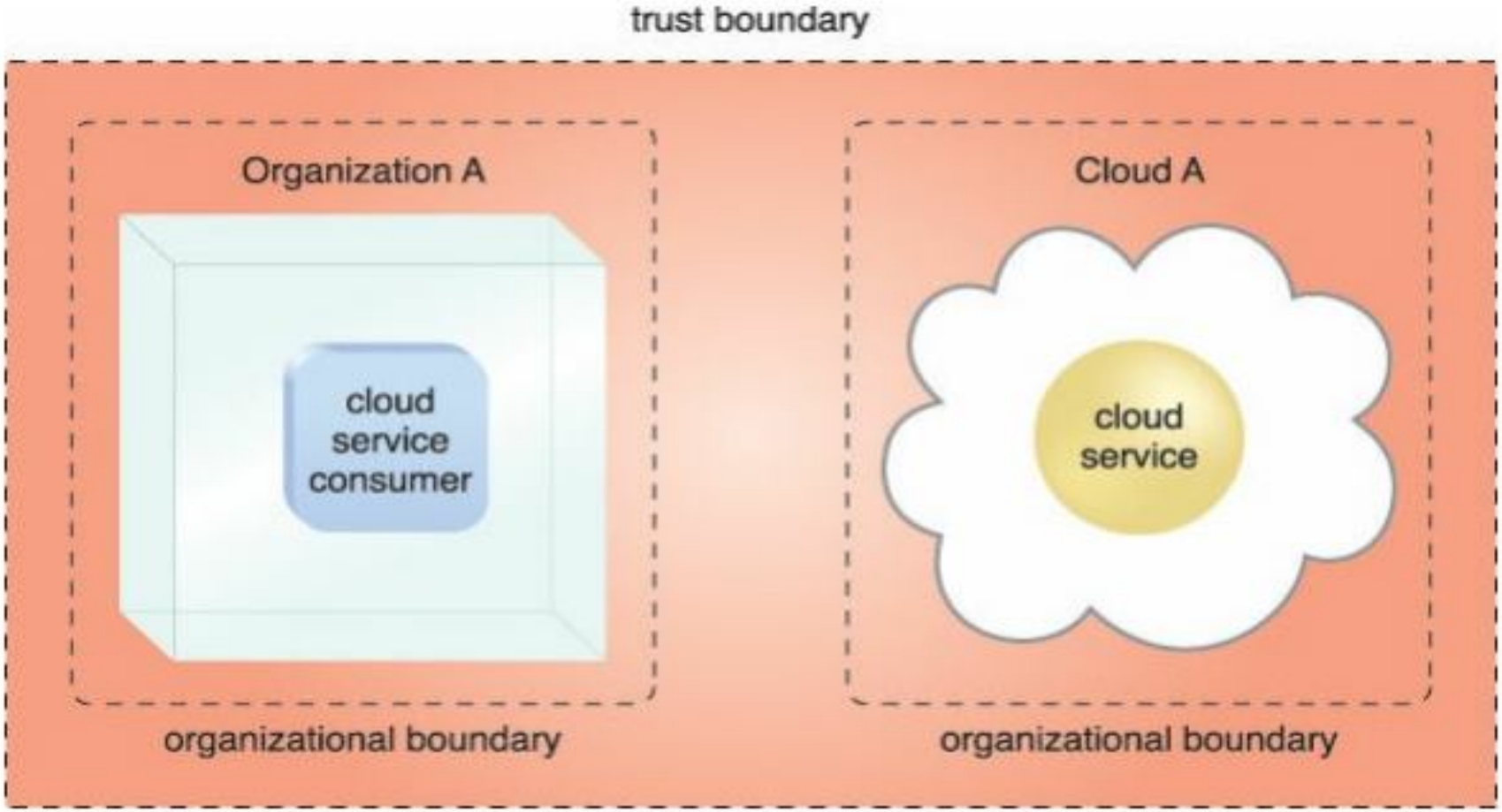
- An organizational boundary represents the physical perimeter that surrounds a set of IT resources that are owned and governed by an organization.
- The organizational boundary does not represent ~~the boundary of an actual organization~~, only an organizational set of IT assets and IT resources.
- Similarly, clouds have an organizational boundary



Organizational boundaries of a cloud consumer (left), and a cloud provider (right), represented by a broken line notation.

Trust Boundary

- When an organization assumes the role of cloud consumer to access cloud-based IT resources, it needs to extend its trust beyond the physical boundary of the organization to include parts of the cloud environment.
- A trust boundary is a logical perimeter that typically spans beyond physical boundaries to represent the extent to which IT resources are trusted.
- When analyzing cloud environments, the trust boundary is most frequently associated with the trust issued by the organization acting as the cloud consumer.



An extended trust boundary encompasses the organizational boundaries of the cloud provider and the cloud consumer.

Cloud Characteristics

Cloud Characteristics

- An IT environment requires a specific set of characteristics to enable the remote provisioning of scalable and measured IT resources in an effective manner.
- These characteristics need to exist to a meaningful extent for the IT environment to be considered an effective cloud.
- The following six specific characteristics are common to the majority of cloud environments:
  - ✓ On-demand usage
  - ✓ Ubiquitous access
  - ✓Multitenancy (and resource pooling)
  - ✓Elasticity
  - ✓Measured usage
  - ✓Resiliency

On-Demand Usage

- A cloud consumer can unilaterally access cloud-based IT resources giving the cloud consumer the freedom to self-provision these IT resources.
- A consumer can self provision computing capabilities, such as server time and network storage, as needed automatically without requiring human interaction with each service provider.
- Once configured, usage of the self-provisioned IT resources can be automated, requiring no further human involvement by the cloud consumer or cloud provider.
- This results in an on-demand usage environment. Also known as "on-demand self-service usage,"

Ubiquitous Access

- Ubiquitous access represents the ability for a cloud service to be widely accessible.
- Establishing ubiquitous access for a cloud service can require support for a range of devices, transport protocols, interfaces, and security technologies.
- To enable this level of access generally requires that the cloud service architecture be tailored to the particular needs of different cloud service consumers.
- Broad network access which has Capabilities to be available over the network and accessed through standard mechanisms that promote use by heterogeneous thin or thick client platforms.

Multitenancy and Resource Pooling

- The characteristic of a software program that enables an instance of the program to serve different consumers(tenants) whereby each is isolated from the other, is referred to as multitenancy.
- A cloud provider pools its IT resources to serve multiple cloud service consumers by using multitenancy models that frequently rely on the use of virtualization technologies.
- Through the use of multitenancy technology, IT resources can be dynamically assigned and reassigned, according to cloud service consumer demands.

Multitenancy and Resource Pooling

- Resource pooling allows cloud providers to pool large-scale IT resources to serve multiple cloud consumers.
- Different physical and virtual IT resources are dynamically assigned and reassigned according to cloud consumer demand, typically followed by execution through statistical multiplexing.
- Resource pooling is commonly achieved through multitenancy technology, and therefore encompassed by this multitenancy characteristic.

Rapid Elasticity

- Elasticity is the automated ability of a cloud to transparently scale IT resources, as required in response to runtime conditions or as predetermined by the cloud consumer or cloud provider.
- Elasticity is often considered a core justification for the adoption of cloud computing, primarily due to the fact that it is closely associated with the Reduced Investment and Proportional Costs benefit.
- Cloud providers with vast IT resources can offer the greatest range of elasticity.

Measured Usage

- It represents the ability of a cloud platform to keep track of the usage of its IT resources, primarily by cloud consumers.
- Based on what is measured, the cloud provider can charge a cloud consumer only for the IT resources actually used and/or for the time frame during which access to the IT resources was granted.
- In this context, measured usage is closely related to the on-demand characteristic.
- Measured usage is not limited to tracking statistics for billing purposes. It also encompasses the general monitoring of IT resources and related usage reporting (for both cloud provider and cloud consumers).

Resiliency

- Resilient computing is a form of failover that distributes redundant implementations of IT resources across physical locations.
- IT resources can be preconfigured so that if one becomes deficient, processing is automatically handed over to another redundant implementation.
- Within cloud computing, the characteristic of resiliency can refer to redundant IT resources within the same cloud (but in different physical locations) or across multiple clouds.
- Cloud consumers can increase both the reliability and availability of their applications by leveraging the resiliency of cloud-based IT resources

Cloud Delivery Models

Cloud Delivery Models

- A cloud delivery model represents a specific, pre-packaged combination of IT resources offered by a cloud provider.
- There are three common cloud delivery models that have become widely established and formalized:
  - ✓ Infrastructure-as-a-Service (IaaS)
  - ✓ Platform-as-a-Service (PaaS)
  - ✓ Software-as-a-Service (SaaS)

Cloud Delivery Models

- Many specialized variations of the three base cloud delivery models have emerged, each comprised of a distinct combination of IT resources.
- Some examples include:
  - ✓ Storage-as-a-Service
  - ✓ Database-as-a-Service
  - ✓ Security-as-a-Service
  - ✓ Communication-as-a-Service
  - ✓ Integration-as-a-Service
  - ✓ Testing-as-a-Service
  - ✓ Process-as-a-Service

Infrastructure-as-a-Service (IaaS)

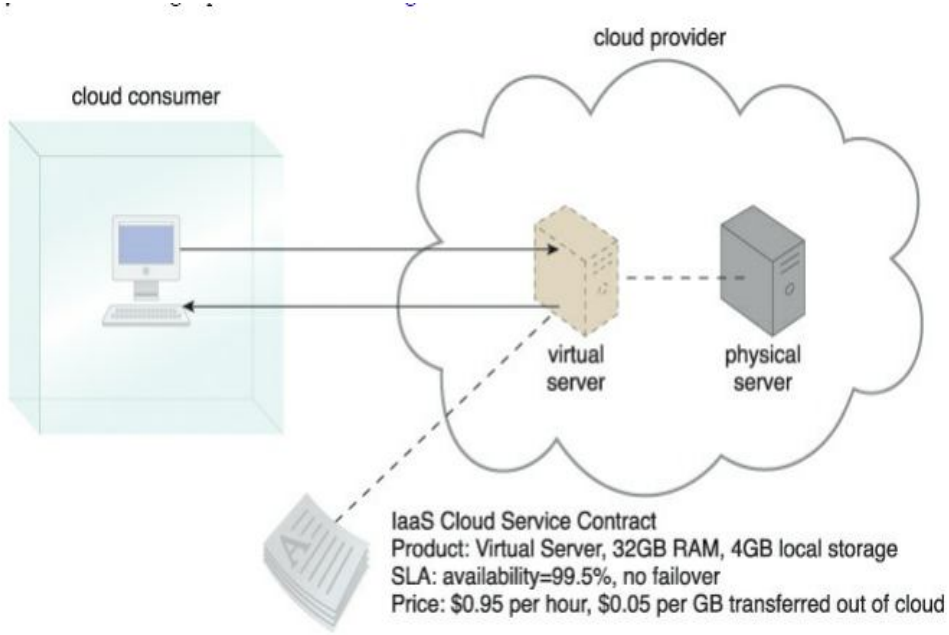
- The IaaS delivery model represents a self-contained IT environment comprised of infrastructure-centric IT resources that can be accessed and managed via cloud service-based interfaces and tools.
- This environment can include hardware, network, connectivity, operating systems, and other "raw" IT resources.
- In contrast to traditional hosting or outsourcing environments, with IaaS, IT resources are typically virtualized and packaged into bundles that simplify up-front runtime scaling and customization of the infrastructure.
- The general purpose of an IaaS environment is to provide cloud consumers with a high level of control and responsibility over its configuration and utilization.

Infrastructure-as-a-Service (IaaS)

- The IT resources provided by IaaS are generally not preconfigured, placing the administrative responsibility directly upon the cloud consumer.
- This model is therefore used by cloud consumers that require a high level of control over the cloud-based environment they intend to create.
- Sometimes cloud providers will contract IaaS offerings from other cloud providers in order to scale their own cloud environments.
- The types and brands of the IT resources provided by IaaS products offered by different cloud providers can vary.

Infrastructure-as-a-Service (IaaS)

- IT resources available through IaaS environments are generally offered as freshly initialized virtual instances.
- A central and primary IT resource within a typical IaaS environment is the virtual server.
- Virtual servers are leased by specifying server hardware requirements, such as processor capacity, memory, and local storage space.



A cloud consumer is using a virtual server within an IaaS environment. Cloud consumers are provided with a range of contractual guarantees by the cloud provider, pertaining to characteristics such as capacity, performance, and availability.

Platform-as-a-Service (PaaS)

- The PaaS delivery model represents a pre-defined "ready-to-use" environment typically comprised of already deployed and configured IT resources.
- Specifically, PaaS relies on (and is primarily defined by) the usage of a ready-made environment that establishes a set of prepackaged products and tools used to support the entire delivery lifecycle of custom applications.
- By working within a ready-made platform, the cloud consumer is spared the administrative burden of setting up and maintaining the bare infrastructure IT resources provided via the IaaS model.
- Conversely, the cloud consumer is granted a lower level of control over the underlying IT resources that host and provision the platform.

Platform-as-a-Service (PaaS)

- The capability provided to the consumer is to deploy onto the cloud infrastructure consumer-created or acquired applications created using programming languages, libraries, services, and tools supported by the provider.
- The consumer does not manage or control the underlying cloud infrastructure including network, servers, operating systems, or storage, but has control over the deployed applications and possibly configuration settings for the application-hosting environment.
- PaaS products are available with different development stacks. *For example, Google App Engine offers a Java and Python-based environment.*

Platform-as-a-Service (PaaS)

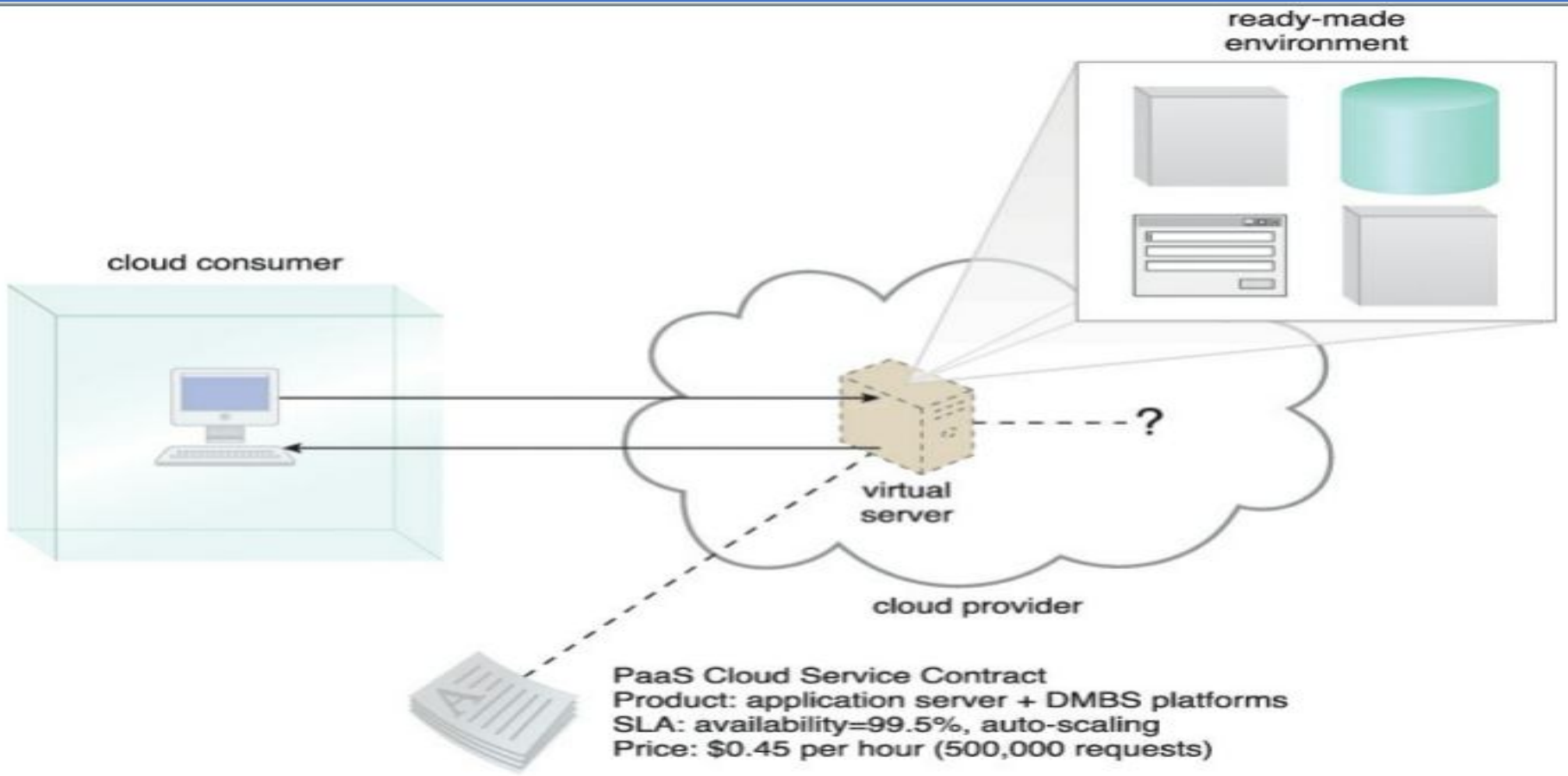


Figure A cloud consumer is accessing a ready-made PaaS environment. The question mark indicates that the cloud consumer is intentionally shielded from the implementation details of the platform.

Software-as-a-Service (SaaS)

- A software program positioned as a shared cloud service and made available as a "product" or generic utility represents the typical profile of a SaaS offering.
- The SaaS delivery model is typically used to make a reusable cloud service widely available (often commercially) to a range of cloud consumers.
- A cloud consumer is generally granted very limited administrative control over a SaaS implementation.

Software-as-a-Service (SaaS)

- It is most often provisioned by the cloud provider, but it can be legally owned by whichever entity assumes the cloud service owner role.
- *For example, an organization acting as a cloud consumer while using and working with a PaaS environment can build a cloud service that it decides to deploy in that same environment as a SaaS offering.*
- The same organization then effectively assumes the cloud provider role as the SaaS-based cloud service is made available to other organizations that act as cloud consumers when using that cloud service.

A comparison of typical cloud delivery model control levels

Cloud Delivery Model	Typical Level of Control Granted to Cloud Consumer	Typical Functionality Made Available to Cloud Consumer
SaaS	usage and usage-related configuration	access to front-end user-interface
PaaS	limited administrative	moderate level of administrative control over IT resources relevant to cloud consumer's usage of platform
IaaS	full administrative	full access to virtualized infra-structure-related IT resources and, possibly, to underlying physical IT resources

Typical activities carried out by cloud consumers and cloud providers

Cloud Delivery Model	Common Cloud Consumer Activities	Common Cloud Provider Activities
SaaS	uses and configures cloud service	implements, manages, and maintains cloud service monitors usage by cloud consumers
PaaS	develops, tests, deploys, and manages cloud services and cloud-based solutions	pre-configures platform and provisions underlying infrastructure, middleware, and other needed IT resources, as necessary monitors usage by cloud consumers
IaaS	sets up and configures bare infrastructure, and installs, manages, and monitors any needed software	provisions and manages the physical processing, storage, networking, and hosting required monitors usage by cloud consumers

Cloud Deployment Models

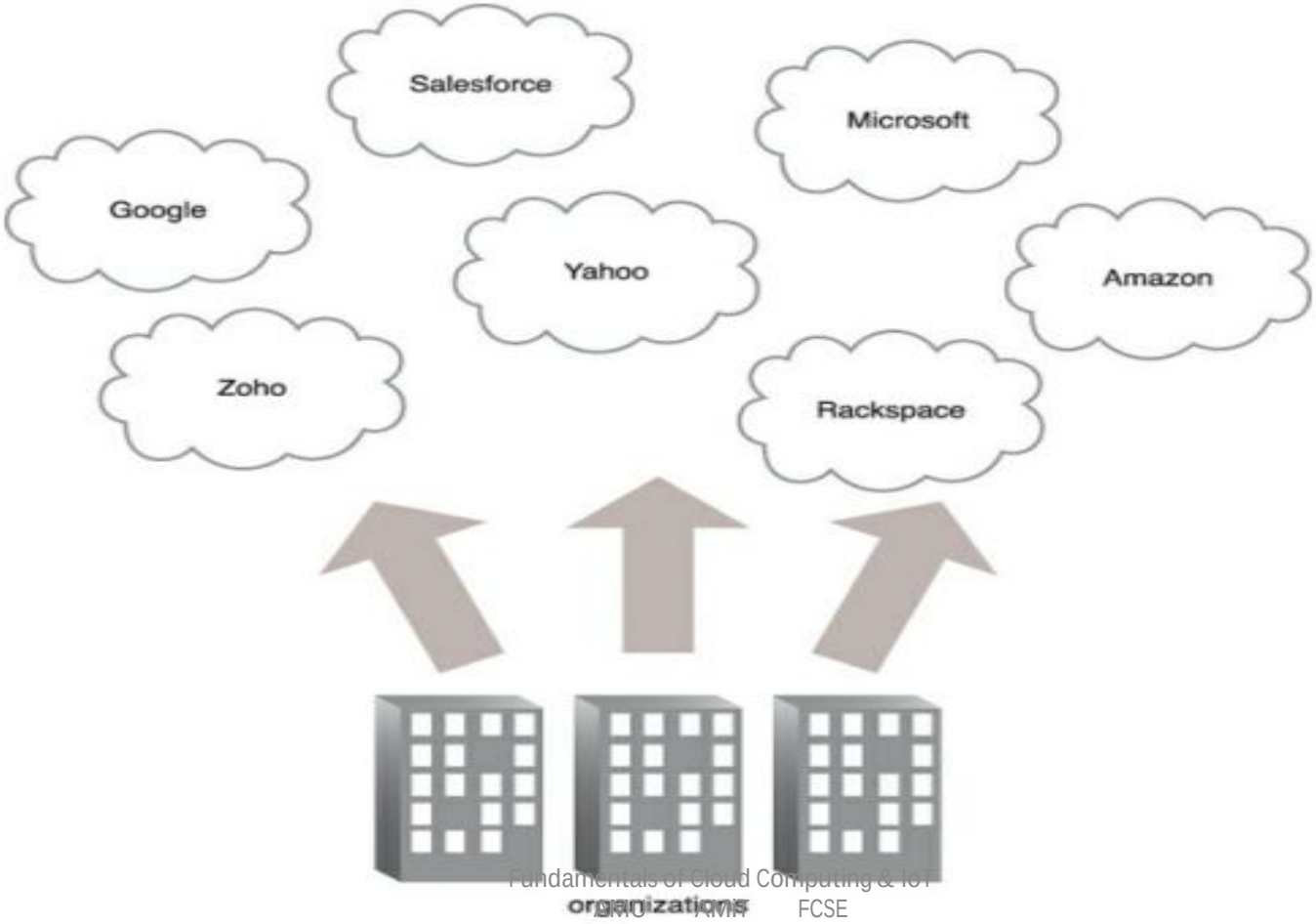
Cloud Deployment Models

- A cloud deployment model represents a specific type of cloud environment, primarily distinguished by ownership, size, and access.
- There are four common cloud deployment models:
  - ✓ Public cloud
  - ✓ Community cloud
  - ✓ Private cloud
  - ✓ Hybrid cloud

Public Clouds

- A public cloud is a publicly accessible cloud environment owned by a third-party cloud provider.
- The IT resources on public clouds are usually provisioned via the previously described cloud delivery models and are generally offered to cloud consumers at a cost or are commercialized via other avenues (such as advertisement).
- The cloud provider is responsible for the creation and on-going maintenance of the public cloud and its IT resources.

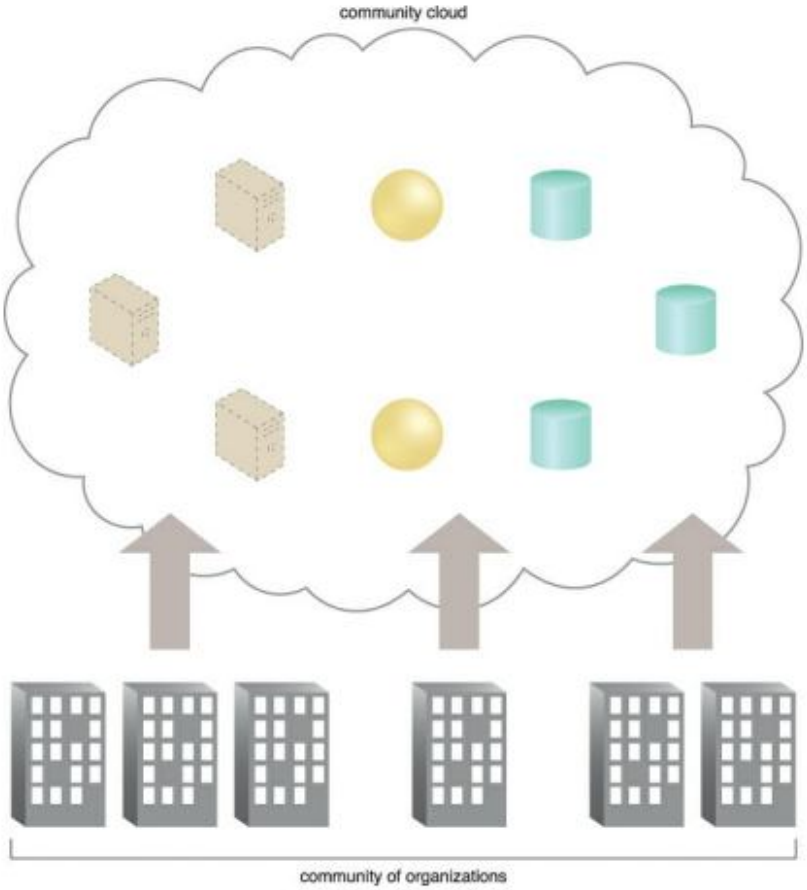
Public Clouds



Community Clouds

- A community cloud is similar to a public cloud except that its access is limited to a specific community of cloud consumers.
- The community cloud may be jointly owned by the community members or by a third-party cloud provider that provisions a public cloud with limited access.
- The member cloud consumers of the community typically share the responsibility for defining and evolving the community cloud.
- Membership in the community does not necessarily guarantee access to or control of all the cloud's IT resources.
- Parties outside the community are generally not granted access unless allowed by the community.

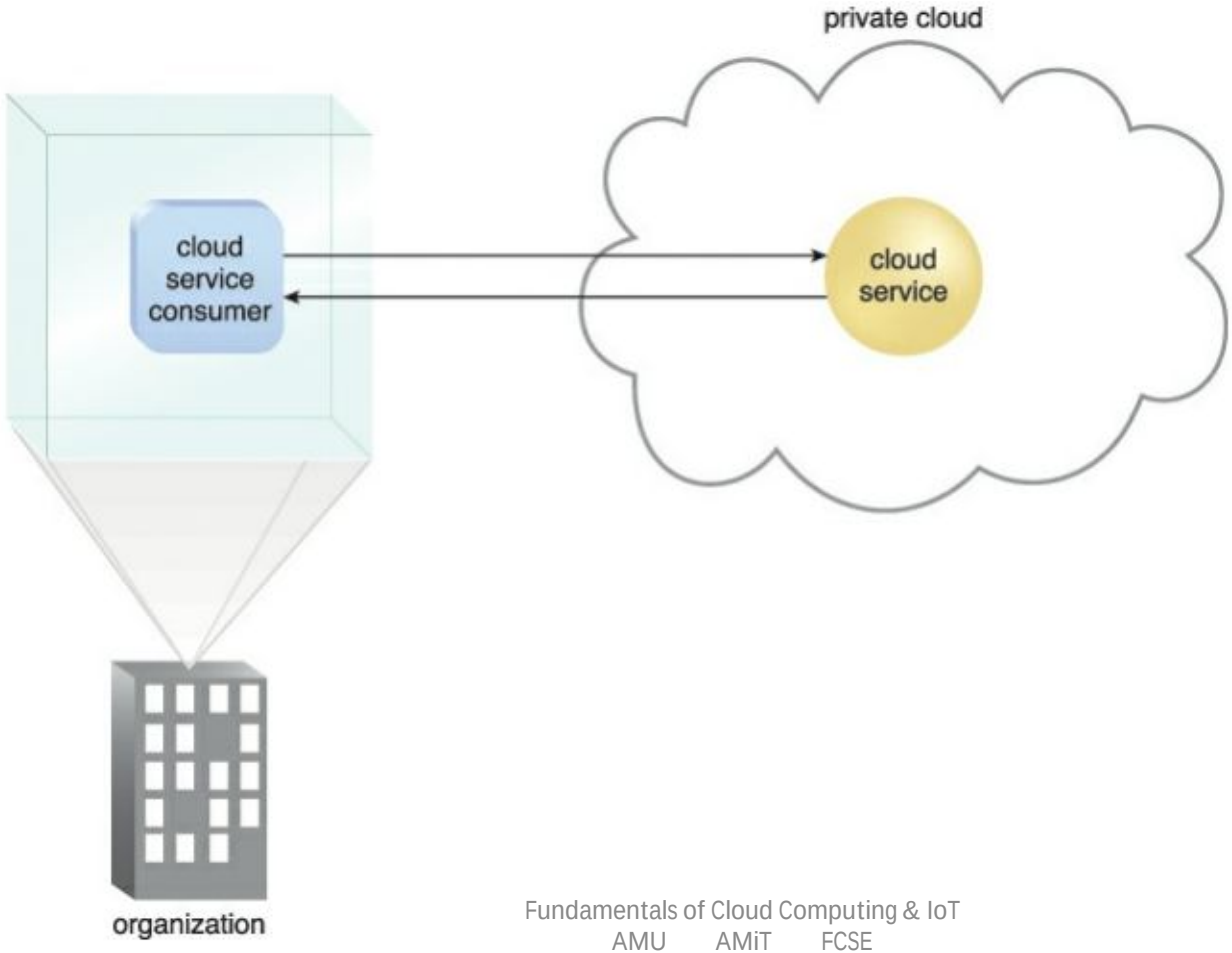
Community Clouds



Private Clouds

- A private cloud is owned by a single organization.
- Private clouds enable an organization to use cloud computing technology as a means of centralizing access to IT resources by different parts, locations, or departments of the organization.
- The use of a private cloud can change how organizational and trust boundaries are defined and applied.
- The actual administration of a private cloud environment may be carried out by internal or outsourced staff. With a private cloud, the same organization is technically both the cloud consumer and cloud provider.

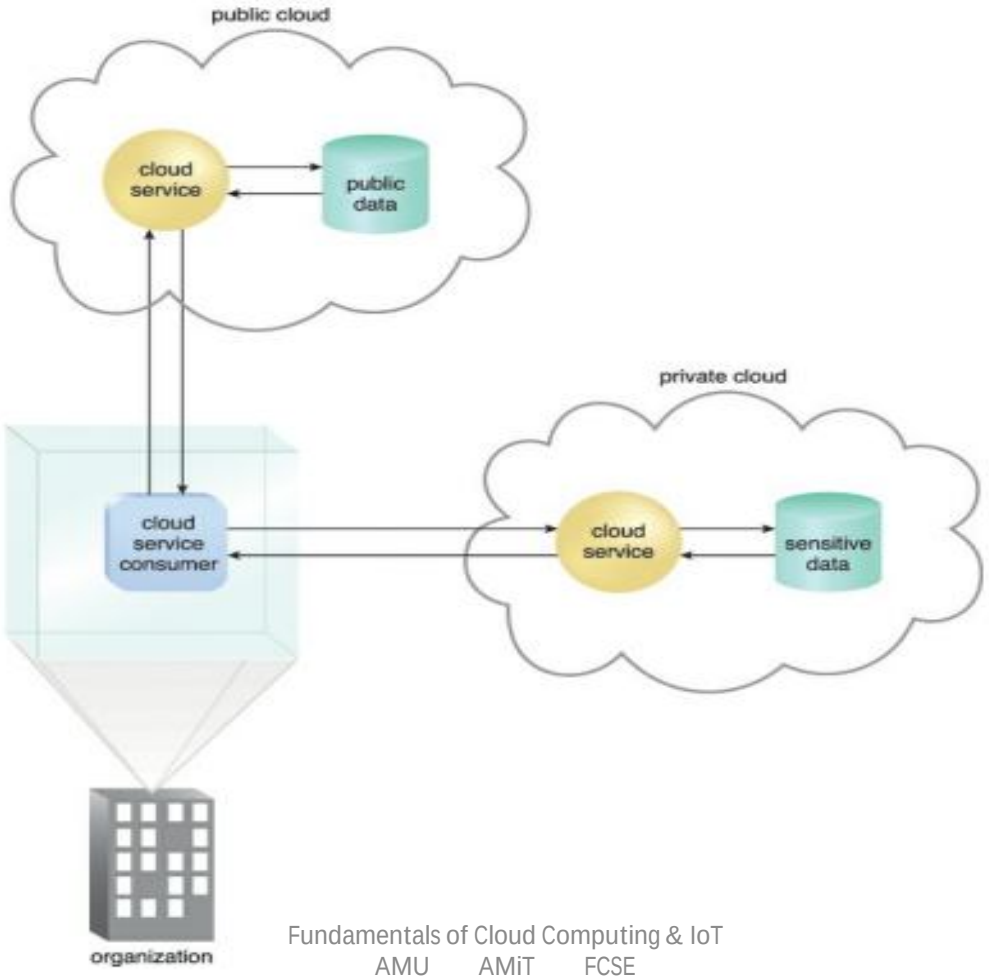
Private Clouds



Hybrid **Clouds**

- A hybrid cloud is a cloud environment comprised of two or more different cloud deployment models.
- *For example, a cloud consumer may choose to deploy cloud services processing sensitive data to a private cloud and other, less sensitive cloud services to a public cloud.*
- The result of this combination is a hybrid deployment model
- Hybrid deployment architectures can be complex and challenging to create and maintain due to the potential disparity in cloud environments and the fact that management responsibilities are typically split between the private cloud provider organization and the public cloud provider.

Hybrid **Clouds**





End of chapter 3

Qn. if Any???



Seminar Titles for Assignments(20%)

1. Edge computing
2. Cloud and E-commerce
3. Pervasive computing
4. Cloud Simulation Tools
5. Cloud Analytics
6. Cluster computing
7. Cloud Carriers; Ethiotelecom vs. Safaricom
8. Cellular Mobile Technologies
9. AI, AR, VR and the Clouds
10. Fog computing
11. Virtualization in Cloud Computing
12. Cloud and E-Governance
13. Grid Computing
14. Load Balancing in Cloud Computing
15. Green computing
16. Cyber security and Cloud SLA
17. Blockchain Technology
18. Big Data Technology
19. Digital Forensics and Cyber Crime
20. Autonomic Computing